

Marissa J. Maroni

(978) 320-9700 | maroni.marissa@gmail.com | www.linkedin.com/in/marissa-maroni/

EDUCATION

University of Pennsylvania, Philadelphia, PA Expected Spring 2025

Ph.D. in Neuroscience – GPA: 4.0

Advisor: Erica Korb, coursework completed Fall 2021

Teaching Certificate, Center for Excellence in Teaching, Learning & Innovation; completed 2024

Bridgewater State University, Bridgewater, MA 2019

B.S. in Biological Sciences with minors in biochemistry and chemistry – GPA: 3.96

Summa cum laude, Departmental Honors

RESEARCH EXPERIENCE

PhD Candidate Fall 2020 – Present

Thesis advisor: Erica Korb Ph.D., University of Pennsylvania, Philadelphia, PA

- Assembled a human patient cohort utilizing international clinical collaborations from 9 countries to identify a neurodevelopmental disorder that arises in individuals with variants in the epigenetic regulator, DOT1L
- Demonstrated using Next-Generation sequencing and behavioral paradigms in mice that DOT1L loss causes transcriptional and behavioral disruptions in mice
- Leveraged primary embryonic neurons and embryonic stem cell-derived neuronal cultures to understand the regulation of the histone variant H3.3 by DOT1L and its posttranslational modification it deposits
- Mentored one undergraduate student and one rotation graduate student
- Communicated research at three international conferences and research resulted in two publications (in progress)

Undergraduate Research Student 2016 – 2019

Thesis advisor: Joseph Seggio Ph.D., Bridgewater State University, Bridgewater, MA

- Honors thesis title: *PYY₃₋₃₆ Efficacy is Independent of Photoperiod but Dependent on Time of Day*
- Studied how disruptions to circadian rhythm and diet impact physiology and behavior using mouse and fly model systems
- Communicated research at two international conferences and research resulted in five publications

Amgen Scholar Summer 2018

P.I.: Miriam B. Goodman Ph.D., Stanford University, Stanford, CA

- Project title: *Mechanosensitive ion channel distributions and its role in touch sensation*
- Utilized the worm, *C. elegans*, to investigate the molecular mechanisms that transduce touch sensation

NSF funded REU Intern Summer 2017

P.I.: Brock Grill Ph.D., The Scripps Research Institute, Jupiter, FL

- Project title: *Investigating the TGF- β /unc-129 Mutant Morphology Defect of the Axon*
- Utilized the worm, *C. elegans*, to study how proteins involved in signaling aid in axon guidance
- Research resulted in one publication

TEACHING & MENTORSHIP EXPERIENCE

Co-director Summer 2023/Summer 2024

Volunteer Teacher Summer 2022

Program: Upward Bound, University of Pennsylvania

- **Knowledge areas in course:** electrical signaling, neuronal circuits, sensory systems (taste and vision), brain development, learning and memory, neuroanatomy, reward system
- Coordinated approximately 10 graduate-level teaching volunteers to team-teach a summer long neuroscience course for high school students

- Taught 5 classes to approximately 12 students and provided individualized support to every class when serving as co-director (8 classes/summer)
- Designed, developed, edited, and customized interactive high school level neuroscience labs that cover course knowledge areas

Teaching Assistant Trainer

Summer 2024

Program: Center for Teaching, Learning, and Innovation, University of Pennsylvania

- **Knowledge areas in course:** constructive and reflective grading, effective time management as a teaching assistant, managing student needs
- Designed an interactive workshop on grading utilizing small group and discussion-based activities
- Led two workshop sessions to 50 graduate-level students
- Advised and led discussion-based sessions on teaching scenarios that cover course knowledge areas

Teaching Assistant

Spring 2022/Spring 2023

Course: Cell Biology and Biochemistry, University of Pennsylvania

- **Knowledge areas in course:** molecular signal transduction, cellular functions (growth, metabolism, motility, and death), cytoskeleton structure
- Led classes dedicated to working through problem sets in small group with 40 graduate-level students
- Created and taught a workshop session about molecular signal transduction to 60 graduate-level students
- Composed and edited experimental-based exam questions along with grading and providing feedback for 120 student exams

Teaching Assistant

Spring 2022/Spring 2023

Course: Introduction to Brain and Behavior, University of Pennsylvania

- **Knowledge areas in course:** neuron signaling (i.e. membrane potential, neurotransmitters), sensory and motor systems, learning and memory
- Taught weekly lecture-style recitations reviewing course materials with 20 undergraduate-level students
- Composed and edited multiple choice and short answer exam questions along with grading tests for 150 students

Graduate Assistant

Summer 2020/Summer 2021

Program: Penn Rising Senior Summer Academy, University of Pennsylvania

- **Knowledge areas in course:** college preparation, financial planning for higher education, resume and college essay construction
- Led daily virtual classes of approximately 15 high school students
- Mentored students in class on college essay writing, resume development, and career exploration
- Developed interactive virtual activities focused on college preparation

Teaching Assistant

Spring 2017 – Spring 2019

Program: Center for the Advancement of STEM Education, Bridgewater State University

- **Knowledge areas in course:** environmental pollutants, water-testing, DNA technologies
- Set-up and helped run daily lab activities of approximately 25 students
- Provided one-one aid to elementary and middle school level students

Undergraduate Mentor

Spring 2022 – Spring 2024

Student: Katherine Lynch, University of Pennsylvania

- Taught techniques including Western blotting, cell culture, immunocytochemistry, RT-qPCR, image analysis, and behavioral analysis
- Advised on project presentations, posters, and master's thesis
- Research resulted in co-authorship on submitted paper at Nature Communications

Graduate Rotation Student Mentor

Student: Annastelle Cohen, University of Pennsylvania

- Taught techniques including Western blotting, transfections, cell culture, nuclei isolation and site-directed mutagenesis
- Rotation work resulted in co-authorship on submitted paper at Nature Communications

SCIENCE OUTREACH

Scientific Writer, Penn Neuro Know, University of Pennsylvania Spring 2021 – Fall 2023

- Wrote, edited, and published 12 articles breaking down neuroscience topics for the Penn Neuro Know blog, focused on condensing neuroscience topics in a language that is understandable to the everyday reader

Board Member, Elementary School Outreach, University of Pennsylvania Spring 2022 – Spring 2023

- Created lessons plans and administered neuroscience lessons to elementary-level students

Mentor, Project SHORT Summer 2020 – Present

- Edited personal statements, advised on the graduate application process, and met with prospective graduate students about graduate admissions

Volunteer, Philadelphia Brain Bee, University of Pennsylvania Spring 2020/Spring 2021/Spring 2023

- Created questions and presented neuroscience-based quiz questions to test high school students' neuroscience knowledge

Volunteer, Science Education Academy, Philadelphia, PA Fall 2019 – Spring 2020

- Administered biology lessons to elementary-level students utilizing hands-on activities

PUBLICATIONS

* Indicates equal contribution

Maroni, M. J., et al. Korb, E. Loss of DOT1L function disrupts neuronal transcription, animal behavior, and leads to a novel neurodevelopmental disorder. *Submitted at Nature Communications*.

Krick, K. S., **Maroni, M. J.**, Korb, E., Lynch K. W., Heller, E. A. Meta-analysis of activated neurons reveals dynamic regulation of diverse classes of alternative splicing. *Under review at Genome Biology*.

Maroni, M. J., Capri, K. M., Cushman, A. V., Deane, H. V., Concepcion, H., DeCoursey, H., & Seggio, J. A. (2020). The timing of fasting leads to different levels of food consumption and PYY 3–36 in nocturnal mice. *Hormones*, 1-10. <https://doi.org/10.1007/s42000-020-00221-x>

Maroni, M. J.*, Capri, K. M.*, Arruda, N. L.*, Gelineau, R. R.*, Deane, H. V., Concepcion, H. A., ... & Logan, R. W. (2020). Substrain specific behavioral responses in male C57BL/6N and C57BL/6J mice to a shortened 21-hour day and high-fat diet. *Chronobiology International*, 37(6), 809-823. <https://doi.org/10.1080/07420528.2020.1756840>

Capri, K. M.*, **Maroni, M. J.***, Deane, H. V., Pierre, A., Adams, A. M., Goncalves, F. L., ... & Seggio, J. A. (2020). Effects of time of day and constant light on the behavioral responses and ethanol metabolism to acute alcohol administration in male Black Swiss mice. *Biological Rhythm Research*, 51(4), 566-585. <https://doi.org/10.1080/09291016.2018.1543640>

Giles, A. C., Desbois, M., Opperman, K. J., Tavora, R., **Maroni, M. J.**, & Grill, B. (2019). A complex containing the O-GlcNAc transferase OGT-1 and the ubiquitin ligase EEL-1 regulates GABA neuron function. *Journal of Biological Chemistry*, 294(17), 6843-6856. <https://doi.org/10.1074/jbc.ra119.007406>

Capri, K. M.*, **Maroni, M. J.***, Deane, H. V., Concepcion, H. A., Decoursey, H., Logan, R. W., & Seggio, J. A. (2019). Male C57BL/6N and C57BL/6J mice respond differently to constant light and running-wheel access. *Frontiers in Behavioral Neuroscience*, 13, 268. <https://doi.org/10.3389/fnbeh.2019.00268>

- Selected for Individual and Social Behaviors – Editor's Pick 2021

Maroni, M. J.*, Capri, K. M.*, Cushman, A. V., Monteiro De Pina, I. K., Chasse, M. H., & Seggio, J. A. (2018). Constant light alters serum hormone levels related to thyroid function in male CD-1 mice. *Chronobiology international*, 35(10), 1456-1463. <https://doi.org/10.1080/07420528.2018.1488259>

PUBLISHED ABSTRACTS

- Maroni, M. J.**, Bartoni, M., Thudium, S. Lynch, K., Lee, R., Deshwar, A., Mehta, S., Quaye, C., Armache, K.J., Costain, G., Korb, E. Loss of Dot1L and H3K79me Impacts Memory, Sociability, and Synaptic Gene Expression in Mice. Society for Neuroscience, Washington D.C. 2023
- Maroni, M. J.**, Wolk, D. A., Das, S. R., De Flores, R., Wisse, L., Xie, L., Yushkevich, P. A., Lee, E. B., and McMillan, C. T. Epigenetic Measurement of Biological Age Associates with Tau Load in Normal Brain Aging. Alzheimer's Association International Conference, Amsterdam, Netherlands (moved to virtual) 2020
- Maroni, M. J.**, Carpi, K. M., Cushman, A. V., Monteiro De Pina, I. K. Chasse, M. H., Seggio, J. A. The physiological effects of PYY is dependent upon time of day. Society for Neuroscience, San Diego, CA 2018
- Carpi, K. M., **Maroni, M. J.**, Chasse, M. H., Cushman, A. V., Seggio, J. A. Day-night variation in ethanol related behaviors in Drosophila and mice. Society for Neuroscience, San Diego, CA 2018
- Maroni, M. J.** Mechanosensitive ion channel distribution and its role in touch sensation. Stanford Summer Research Program Symposium, Stanford, CA, 2018
- Cushman, A. V., Arruda, N. L., Gelineau, R. R., Monteiro De Pina, I. K., **Maroni, M. J.**, Capri, K. M., Seggio, J. A. Effects of lithium and high fat diet on sociability and anxiety in Black Swiss mice. Society for Neuroscience, Washington D.C., 2017

AWARDS & HONORS

F31 National Research Service Award; \$53k/2.5 years	2023 – 2025
NSF Graduate Research Fellowship Honorable Mention	2021
Dr. Madhu N. Rao Scholarship; \$1,000	2019
NSF Graduate Research Fellowship Honorable Mention	2019
Bridgewater's Best, awarded to 5 seniors with a record of service through leadership	2019
Adrian Tinsley Semester Research Grant; \$1,800	2016 – 2019
Amgen Scholar – Stanford Summer Research Program (SSRP); \$3,600	2018
John and Abigail Adams Scholarship; \$3,640	2015 – 2019
Goldwater Scholarship Honorable Mention	2018
Bridgewater Alumni Association Scholarship; \$1,500	2018
Patricia Bartlett '67 Endowed Scholarship; \$5,000	2017
Introductory Chemistry Achievement Award	2016

POSTER PRESENTATIONS

1. Loss of Dot1L and H3K79me impacts long-term memory and synaptic gene expression in neurons. Society for Neuroscience, Washington, D.C. Fall 2023.
2. Loss of Dot1L and H3K79me impacts long-term memory and synaptic gene expression in neurons. 3rd Neuroepigenetics & Neuroepitranscriptomics Conference, Riviera Maya, MX . Spring 2023.
3. Loss of Dot1L and H3K79me impacts long-term memory and synaptic gene expression in neurons. Epigenetics Symposium, University of Pennsylvania. Winter 2023.
4. Epigenetic Measurement of Biological Age Associates with Tau Load in Normal Brain Aging. Alzheimer's Association International Conference (virtual). Summer 2020.
5. Investigating the Influence of Epigenetic Age on Aging Signatures. Center for Neurodegenerative Disease Research Retreat, University of Pennsylvania. Fall 2019.
6. Mechanosensitive ion channel distributions and its role in touch sensation. Stanford Summer Research Program Symposium, Stanford University. Summer 2018.
7. The Physiological Effects of PYY is Dependent upon Time of Day. Society for Neuroscience, San Diego, CA. Fall 2018.

8. The Physiological Effects of PYY is Dependent upon Time of Day. Society for Research on Biological Rhythms, Amelia Island, FL. Spring 2018.
9. Investigating the TGF- β /unc-129 Mutant Morphology Defect of the Axon. Undergraduate Summer Internship Research Poster Presentations, The Scripps Research Institute. Summer 2017.

ORAL PRESENTATIONS

1. The role of H3K79 methylation in intellectual disability and H3.3 exchange. H3.3 Symposium, University of Pennsylvania. Spring 2021.
2. PYY₃₋₃₆ Efficacy is Independent of Photoperiod but Dependent on Time of Day. Honors Thesis Defense, Bridgewater State University. Spring 2019.
3. Mechanosensitive ion channel distributions and its role in touch sensation. Friday Informal Seminar Hour (FISH), Bridgewater State University. Fall 2018.
4. Mechanosensitive ion channel distributions and its role in touch sensation. Stanford Summer Research Program Symposium, Stanford University. Summer 2018.
5. Investigating Axon Morphology Defects in *C. elegans* and Creating a Plasmid Library. Friday Informal Seminar Hour (FISH), Bridgewater State University. Fall 2017

SKILLS

Teaching: developing interactive lab activities, differentiated instruction including lecture-style and small group classrooms, mentoring and providing reflective feedback for students and educators, lesson planning, developing course material and content

Cell culture: primary neuronal cultures, patient-derived fibroblasts, mouse embryonic stem cells, cell lines (HEK293T and Neuro-2A), stem cell to neuron differentiation, lentiviral production, transfection

Cell and molecular biology: site-directed mutagenesis, histone extraction, RT-qPCR, Western blotting, immunocytochemistry, confocal imaging, bulk-RNA-sequencing, nuclei isolation

Animals: maintenance and care of the following model systems (mouse, flies, *C. elegans*, and planarians), behavioral testing in mice, adult and embryonic brain microdissections in mice

Computation: proficient in R, processed and analyzed RNA-sequencing datasets, fluorescence imaging analysis in ImageJ

ORGANIZATIONAL MEMBERSHIPS

Graduate-Lead Initiatives and Activities

Member	2020 – Present
Elementary Outreach Board Member	Spring 2020 – Spring 2022
Upward Bound Co-director	Fall 2022 – Present
Tri Beta Biology Honor Society – Theta Lambda Chapter	Spring 2018 – Present
Member	
Vice President for Student Affairs Student Advisory Board	Fall 2018 – Spring 2019
Member	
Panhellenic Association	Spring 2016 – Spring 2019
President	2018 – 2019
Vice President of Leadership Development – Panhellenic Association	2017 – 2018
Phi Sigma Sigma – Epsilon Iota	Spring 2016 – Spring 2019
Multicultural Chair	2017
Junior Panhellenic Delegate	2016
Ritual Chair	2016